

Linear Representations of Groups

Road map

Representation theory replaces an abstract algebraic object by linear maps on a vector space. Once this is done, one can use matrices, eigenvalues, inner products, tensor products, and invariant subspaces. This is especially important in quantum mechanics, where the state space is already a Hilbert space and symmetries act linearly or unitarily.

In this chapter we shall:

1. define linear and matrix representations;
2. prove carefully how the matrix multiplication law follows from the abstract homomorphism law;
3. introduce faithful, trivial, and unitary representations;
4. work out a two-dimensional representation of the quaternion group;
5. construct scalar, vector, and tensor representations of rotation groups;
6. derive Neumann's principle for material tensors and reproduce the full crystal-system table of independent tensor components;
7. construct infinite-dimensional representations on solution spaces of the Helmholtz and Schrödinger equations.

20.1 From an abstract group to linear maps

Let V be a vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Its general linear group is

$$GL(V) = \{A : V \rightarrow V : A \text{ is linear and bijective}\}.$$

If $V = \mathbb{K}^n$, then

$$GL(V) = GL(n, \mathbb{K}) = \{A \in \text{Mat}(n, \mathbb{K}) : \det A \neq 0\}.$$

Definition 20.1: Linear representation

Let G be a group and V a vector space over $\mathbb{K}$. A *linear representation* of G on V is a group homomorphism

$$D : G \longrightarrow GL(V), \quad a \longmapsto D(a).$$

Thus

$$D(a_1 a_2) = D(a_1) D(a_2) \quad (a_1, a_2 \in G). \quad (20.1)$$

The vector space V is the *representation space*. Its dimension is the dimension of the representation:

$$\dim D := \dim_{\mathbb{K}} V.$$

The homomorphism law immediately implies

$$D(e) = I_V, \quad D(a^{-1}) = D(a)^{-1}.$$

Indeed,

$$D(e) = D(e)^2,$$

and $D(e)$ is invertible, so $D(e) = I_V$. Then

$$D(a) D(a^{-1}) = D(e) = I_V.$$

20.1.1 Linear versus matrix representations

Choose an ordered basis

$$\mathcal{B} = (e_1, \dots, e_n)$$

of an n -dimensional representation space V . For $a \in G$, write

$$D(a)e_j = \sum_{i=1}^n D_{\mathcal{B}}(a)_{ij} e_i. \quad (20.2)$$

The coefficients form an invertible matrix

$$D_{\mathcal{B}}(a) = (D_{\mathcal{B}}(a)_{ij}) \in GL(n, \mathbb{K}).$$

The map

$$D_{\mathcal{B}} : G \longrightarrow GL(n, \mathbb{K})$$

is called the *matrix representation* in the basis $\mathcal{B}$.

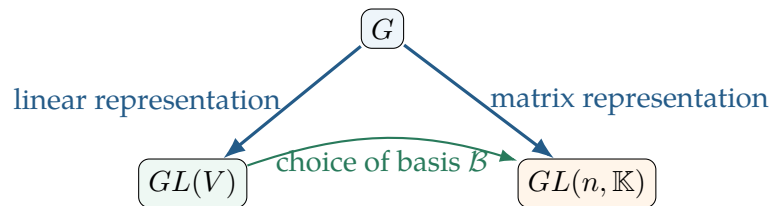


图 20.1: A matrix representation is the coordinate form of a linear representation after choosing a basis.

20.1.2 The multiplication law in indices

Starting from (20.1),

$$D(a)D(b) = D(ab).$$

Apply both sides to a basis vector e_j :

$$\begin{aligned} D(ab)e_j &= \sum_i D_{\mathcal{B}}(ab)_{ij}e_i, \\ D(a)D(b)e_j &= D(a) \left(\sum_k D_{\mathcal{B}}(b)_{kj}e_k \right) \\ &= \sum_k D_{\mathcal{B}}(b)_{kj} D(a)e_k \\ &= \sum_k D_{\mathcal{B}}(b)_{kj} \sum_i D_{\mathcal{B}}(a)_{ik}e_i \\ &= \sum_i \left(\sum_k D_{\mathcal{B}}(a)_{ik} D_{\mathcal{B}}(b)_{kj} \right) e_i. \end{aligned}$$

Uniqueness of the coordinate expansion gives

$$D_{\mathcal{B}}(ab)_{ij} = \sum_k D_{\mathcal{B}}(a)_{ik} D_{\mathcal{B}}(b)_{kj}.$$

This is exactly the matrix multiplication rule

$$D_{\mathcal{B}}(ab) = D_{\mathcal{B}}(a)D_{\mathcal{B}}(b).$$

20.2 A first finite-group example: the quaternion group

The quaternion group is

$$Q_8 = \{\pm 1, \pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\},$$

with defining relations

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1. \quad (20.3)$$

Equivalently,

$$\mathbf{ij} = \mathbf{k}, \quad \mathbf{ji} = -\mathbf{k},$$

and their cyclic analogues.

Define $D : Q_8 \rightarrow GL(2, \mathbb{C})$ on the generators by

$$\begin{aligned} D(1) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & D(-1) &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \\ D(\mathbf{i}) &= \begin{pmatrix} \mathbf{i} & 0 \\ 0 & -\mathbf{i} \end{pmatrix} = \mathbf{i}\sigma_3, & D(\mathbf{j}) &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathbf{i}\sigma_2, \\ D(\mathbf{k}) &= \begin{pmatrix} 0 & \mathbf{i} \\ \mathbf{i} & 0 \end{pmatrix} = \mathbf{i}\sigma_1. \end{aligned} \quad (20.4)$$

The negative elements are represented by

$$D(-q) = -D(q).$$

Let us check the defining relations rather than merely assert them:

$$\begin{aligned} D(\mathbf{i})^2 &= D(\mathbf{j})^2 = D(\mathbf{k})^2 = -I_2 = D(-1), \\ D(\mathbf{i})D(\mathbf{j}) &= \begin{pmatrix} \mathbf{i} & 0 \\ 0 & -\mathbf{i} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{i} \\ \mathbf{i} & 0 \end{pmatrix} = D(\mathbf{k}), \\ D(\mathbf{j})D(\mathbf{i}) &= -D(\mathbf{k}). \end{aligned}$$

Thus every relation in Q_8 is respected, so D is a two-dimensional complex representation.

Defining representations

Matrix groups such as $GL(n, \mathbb{K})$, $O(n)$, $SO(n)$, and $SU(n)$ come with a natural *defining representation* on $\mathbb{K}^n$:

$$D(g)v = gv.$$

An abstract group, by contrast, may possess many inequivalent representations of different dimensions.

20.3 Important adjectives for representations

Definition 20.2: Faithful representation

A representation $D : G \rightarrow GL(V)$ is *faithful* if it is one-to-one, or equivalently if

$$\ker D = \{e\}.$$

A representation may preserve fewer distinctions than the original group. Every element in $\ker D$ acts as the identity on V , so those group elements become invisible in the representation.

Theorem 20.1: A representation is faithful on the quotient by its kernel

For any representation $D : G \rightarrow GL(V)$, the induced map

$$\overline{D} : G / \ker D \longrightarrow GL(V), \quad a \ker D \longmapsto D(a),$$

is a faithful representation of $G / \ker D$.

证明. The kernel of a group homomorphism is normal, so the quotient is defined. If $a \ker D = b \ker D$, then $b^{-1}a \in \ker D$, hence

$$D(a) = D(b)D(b^{-1}a) = D(b),$$

so $\overline{D}$ is well-defined. It is a homomorphism because

$$\overline{D}((a \ker D)(b \ker D)) = D(ab) = D(a)D(b).$$

Finally,

$$\overline{D}(a \ker D) = I_V \implies a \in \ker D \implies a \ker D = \ker D.$$

Thus $\ker \overline{D}$ is trivial. ■

Definition 20.3: Trivial representation

The *trivial* or *unit representation* is

$$D : G \longrightarrow GL(V), \quad D(a) = I_V \quad (a \in G).$$

It deliberately discards all information about the group action.

Definition 20.4: Unitary representation

Let V be a complex inner-product space. A representation

$$D : G \longrightarrow GL(V)$$

is *unitary* if

$$\langle D(a)v_1, D(a)v_2 \rangle = \langle v_1, v_2 \rangle \quad (a \in G, v_1, v_2 \in V). \quad (20.5)$$

Equivalently,

$$D(a)^\dagger = D(a)^{-1} = D(a^{-1}).$$

20.4 Tensor representations of the rotation group

Let $G = SO(3)$, or let G be a subgroup of $SO(3)$ describing the proper rotational symmetries of a material.

20.4.1 Scalars and vectors

A scalar s transforms under the one-dimensional trivial representation:

$$s' = D(R)s = s, \quad D(R) = 1.$$

A vector $\mathbf{v} \in \mathbb{R}^3$ transforms under the defining representation:

$$\mathbf{v}' = D(R)\mathbf{v} = R\mathbf{v},$$

or, in components,

$$v'_i = \sum_{j=1}^3 R_{ij}v_j. \quad (20.6)$$

20.4.2 Rank-two tensors

Let $C = (C_{ij})$ be a rank-two Cartesian tensor. Under a rotation,

$$C'_{ij} = \sum_{k,\ell=1}^3 R_{ik} R_{j\ell} C_{k\ell}. \quad (20.7)$$

In matrix notation,

$$C' = R C R^T. \quad (20.8)$$

The nine-dimensional space of rank-two tensors is

$$\mathbb{R}^3 \otimes \mathbb{R}^3.$$

If the vector representation is $D^{(1)}(R) = R$, then the tensor-product representation is

$$D^{(1)} \otimes D^{(1)} : R \longmapsto R \otimes R.$$

With the double index (i, j) , its matrix entries are

$$D^{(2)}(R)_{(i,j),(k,\ell)} = R_{ik} R_{j\ell}.$$

Equation (20.7) is simply

$$C' = D^{(2)}(R)C$$

after the two indices are flattened into one nine-valued index.

20.5 Linear response and Neumann's principle

Consider Ohm's law in an anisotropic material:

$$j_i = \sum_j \sigma_{ij} E_j, \quad \text{or} \quad \mathbf{j} = \sigma \mathbf{E}. \quad (20.9)$$

Here $\mathbf{j}$ is the current density, $\mathbf{E}$ is the electric field, and σ is the rank-two conductivity tensor.

Rotate the coordinate frame by R . The vector components become

$$\mathbf{j}' = R\mathbf{j}, \quad \mathbf{E}' = R\mathbf{E}.$$

Since $\mathbf{E} = R^T \mathbf{E}'$,

$$\mathbf{j}' = R\sigma\mathbf{E} = R\sigma R^T \mathbf{E}'.$$

Thus the conductivity components in the rotated frame are

$$\boxed{\sigma' = R\sigma R^T, \quad \sigma'_{ij} = \sum_{k,\ell} R_{ik} R_{j\ell} \sigma_{k\ell}.} \quad (20.10)$$

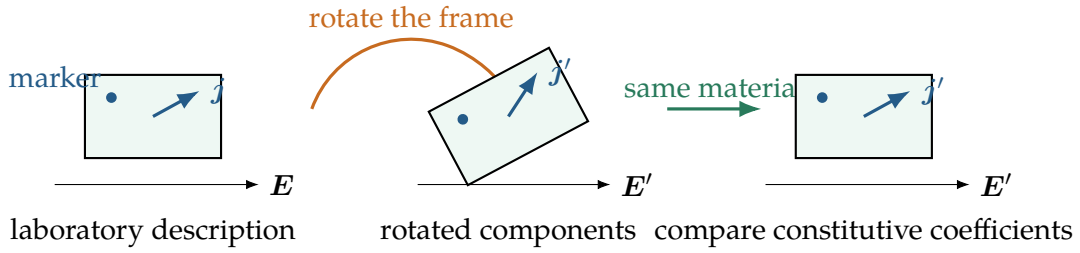


图 20.2: A constitutive law is independent of the coordinate description; the tensor components transform with the rotated basis.

Theorem 20.2: Neumann's principle

Every tensor describing a physical property of a crystal is invariant under every symmetry transformation of that crystal.

For the conductivity tensor and a material symmetry group $G \leq SO(3)$, Neumann's principle says

$$\sigma = R\sigma R^T \quad (R \in G). \quad (20.11)$$

Since $R^{-1} = R^T$, this is equivalent to

$$[R, \sigma] = 0 \quad (R \in G). \quad (20.12)$$

Thus the allowed material tensors form the invariant subspace

$$(\mathbb{R}^3 \otimes \mathbb{R}^3)^G = \{C : D^{(2)}(R)C = C \text{ for every } R \in G\}.$$

For the proper rotational symmetry group $\mathcal{O}$ of a cube, the defining three-dimensional representation is irreducible and its commutant consists only of scalar matrices. Schur's lemma therefore gives

$$\sigma = \sigma_0 I_3. \quad (20.13)$$

The conductivity of a cubic crystal is isotropic even though the crystal itself has only a finite rotation group.

20.5.1 Hooke's law and fourth-rank tensors

In linear elasticity,

$$\tau_{ij} = \sum_{k,\ell} C_{ijkl} \varepsilon_{kl}, \quad (20.14)$$

where τ is the stress tensor, ε is the strain tensor, and C is the rank-four stiffness tensor. Before imposing symmetries, a general rank-four tensor in three dimensions has $3^4 = 81$ components. The physical minor and major symmetries

$$C_{ijkl} = C_{jikl} = C_{ijlk} = C_{klij}$$

reduce this to 21. Crystal symmetries reduce it further. A cubic crystal has only three independent elastic constants, conventionally

$$C_{11}, \quad C_{12}, \quad C_{44}.$$

An isotropic solid has only two, the Lamé parameters.

Bravais system	Point-group symbol		Number of independent components			
	International	Schoenflies	first	second	third	fourth
triclinic	1	C_1	3	6	18	21
	$\bar{1}$	S_2	0	6	0	21
monoclinic	2	C_2	1	4	8	13
	m	C_{1h}	2	4	10	13
	$2/m$	C_{2h}	0	4	0	13
orthorhombic	222	D_2	0	3	3	9
	$mm2$	C_{2v}	1	3	5	9
	mmm	D_{2h}	0	3	0	9
tetragonal	4	C_4	1	2	4	7
	$\bar{4}$	S_4	0	2	4	7
	$4/m$	C_{4h}	0	2	0	7
	422	D_4	0	2	1	6
	$4mm$	C_{4v}	1	2	3	6
	$\bar{4}2m$	D_{2d}	0	2	2	6
	$4/mmm$	D_{4h}	0	2	0	6
rhombohedral	3	C_3	1	2	6	7
	$\bar{3}$	S_6	0	2	0	7
	32	D_3	0	2	2	6
	$3m$	C_{3v}	1	2	4	6
	$\bar{3}m$	D_{3d}	0	2	0	6
hexagonal	6	C_6	1	2	4	5
	$\bar{6}$	C_{3h}	0	2	2	5
	$6/m$	C_{6h}	0	2	0	5
	622	D_6	0	2	1	5
	$6mm$	C_{6v}	1	2	3	5
	$\bar{6}m2$	D_{3h}	0	2	1	5
	$6/mmm$	D_{6h}	0	2	0	5
cubic	23	T	0	1	1	3
	$m\bar{3}$	T_h	0	1	0	3
	432	O	0	1	0	3
	$\bar{4}3m$	T_d	0	1	1	3
	$m\bar{3}m$	O_h	0	1	0	3
isotropic	—	—	0	1	0	2

表 20.1: Number of independent elements of physical tensor properties of orders one through four for the 32 crystallographic point groups. The second-, third-, and fourth-order columns incorporate the conventional internal index symmetries of the corresponding material-property tensors. The data reproduce the table cited in the handwritten notes from J. W. Morris, *Notes on the Thermodynamics of Solids* (2008).

20.6 Infinite-dimensional representations on function spaces

Representations need not be finite-dimensional. Symmetry groups often act naturally on spaces of functions, differential-equation solutions, or quantum wavefunctions.

20.6.1 The Helmholtz equation

Fix $k > 0$, and let

$$\mathcal{S}_k = \{f : \mathbb{R}^3 \rightarrow \mathbb{R} : \Delta f + k^2 f = 0\}. \quad (20.15)$$

Because the Helmholtz equation is linear, $\mathcal{S}_k$ is a vector space, generally infinite-dimensional.

Let $E(3) = \mathbb{R}^3 \rtimes O(3)$ be the Euclidean isometry group. Write an isometry as $g = (\mathbf{a}, R)$, acting by

$$g\mathbf{x} = \mathbf{a} + R\mathbf{x}, \quad g^{-1}\mathbf{x} = R^{-1}(\mathbf{x} - \mathbf{a}).$$

Define the pullback action

$$[D(g)f](\mathbf{x}) := f(g^{-1}\mathbf{x}) = f(R^{-1}(\mathbf{x} - \mathbf{a})). \quad (20.16)$$

The map $D(g) : \mathcal{S}_k \rightarrow \mathcal{S}_k$ is linear:

$$D(g)(z_1 f_1 + z_2 f_2) = z_1 D(g)f_1 + z_2 D(g)f_2.$$

It is a homomorphism because

$$\begin{aligned} [D(g_1)D(g_2)f](\mathbf{x}) &= [D(g_2)f](g_1^{-1}\mathbf{x}) \\ &= f(g_2^{-1}g_1^{-1}\mathbf{x}) \\ &= f((g_1g_2)^{-1}\mathbf{x}) \\ &= [D(g_1g_2)f](\mathbf{x}). \end{aligned}$$

Finally, the Laplacian is invariant under translations and orthogonal transformations:

$$\Delta(f \circ g^{-1}) = (\Delta f) \circ g^{-1}.$$

Therefore

$$(\Delta + k^2)D(g)f = D(g)(\Delta + k^2)f = 0.$$

Hence

$$D : E(3) \longrightarrow GL(\mathcal{S}_k)$$

is an infinite-dimensional representation of the Euclidean group.

20.6.2 Rotationally invariant Schrödinger operators

Let

$$H = -\frac{\hbar^2}{2m}\Delta + V(\mathbf{x}) \quad (20.17)$$

act on $L^2(\mathbb{R}^3)$, and fix an energy E . Its eigenspace is

$$\mathcal{E}_E = \{\psi \in L^2(\mathbb{R}^3) : H\psi = E\psi\}.$$

Assume a subgroup $G \leq SO(3)$ preserves the potential:

$$V(R\mathbf{x}) = V(\mathbf{x}) \quad (R \in G). \quad (20.18)$$

Define

$$[D(R)\psi](\mathbf{x}) = \psi(R^{-1}\mathbf{x}). \quad (20.19)$$

The Laplacian is rotation-invariant and

$$V(\mathbf{x})\psi(R^{-1}\mathbf{x}) = V(R^{-1}\mathbf{x})\psi(R^{-1}\mathbf{x}).$$

Thus

$$\begin{aligned} H[D(R)\psi](\mathbf{x}) &= [H\psi](R^{-1}\mathbf{x}) \\ &= E\psi(R^{-1}\mathbf{x}) = E[D(R)\psi](\mathbf{x}). \end{aligned}$$

Therefore each energy eigenspace is G -invariant, and

$$D : G \longrightarrow GL(\mathcal{E}_E)$$

is a representation.

Its dimension is the degeneracy of the energy E :

$$\dim D = \dim \mathcal{E}_E.$$

The representation is unitary. Indeed, with the usual L^2 inner product and the change of variables $\mathbf{y} = R^{-1}\mathbf{x}$,

$$\begin{aligned} \langle D(R)\psi, D(R)\phi \rangle &= \int_{\mathbb{R}^3} \psi(R^{-1}\mathbf{x})^* \phi(R^{-1}\mathbf{x}) \, d^3x \\ &= \int_{\mathbb{R}^3} \psi(\mathbf{y})^* \phi(\mathbf{y}) \, d^3y \\ &= \langle \psi, \phi \rangle, \end{aligned}$$

because $|\det R| = 1$.

Symmetry and degeneracy

Since H commutes with every $D(R)$,

$$HD(R) = D(R)H,$$

symmetry maps an energy eigenstate to another state of the same energy. Nontrivial representation multiplets therefore produce systematic degeneracies. The precise relation between irreducibility and degeneracy will be developed in the next chapters.

What to retain from this chapter

1. A representation of G on V is a homomorphism

$$D : G \rightarrow GL(V), \quad D(ab) = D(a)D(b).$$

Its dimension is $\dim V$.

2. Choosing a basis converts a linear representation into a matrix representation. A different basis changes all matrices by a simultaneous similarity transformation.
3. A representation is faithful exactly when $\ker D = \{e\}$. In general it induces a faithful representation of $G/\ker D$.

4. A unitary representation preserves inner products and satisfies

$$D(a)^\dagger = D(a)^{-1} = D(a^{-1}).$$

5. Under rotations, scalars transform trivially, vectors by R , and rank-two tensors by

$$C' = RCR^\top, \quad C'_{ij} = R_{ik}R_{j\ell}C_{k\ell}.$$

6. Neumann's principle requires every material-property tensor to be invariant under the material's symmetry group. For conductivity,

$$R\sigma R^\top = \sigma.$$

Symmetry can drastically reduce the number of independent tensor components.

7. Isometries act on functions by pullback,

$$[D(g)f](\mathbf{x}) = f(g^{-1}\mathbf{x}),$$

producing infinite-dimensional representations on invariant solution spaces.

8. If a Schrödinger Hamiltonian is invariant under G , each energy eigenspace is a unitary representation space of G , and its dimension is the degeneracy of that energy.